Sahaj Bhadja

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Summary

Software engineer experienced in **Python, Java, TypeScript, APIs, PostgreSQL, SQL Server, distributed systems, automated testing, monitoring, and observability**, building reliable, scalable software for data-driven operational workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built distributed FastAPI services processing 5,000+ monthly jobs with explicit state transitions, failure isolation, validation, and reconciliation, increasing concurrent throughput 3x while reliably sustaining 99%+ production reliability.
- Optimized PostgreSQL and SQL Server schemas, indexing, caching, and access patterns for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds and integration latency 35%.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented Python services and REST APIs processing 1,000+ files per batch, adding multithreading, fault-tolerant execution, diagnostics, fallback paths, and automated tests for reliable peak-load performance at scale.
- Troubleshot production issues across 10,000+ daily requests, resolving OCR, model, ingestion, and API failures while collaborating through code reviews to improve maintainability and reduce feedback turnaround 60%.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Developed Python and PyTorch components processing 100,000+ trajectory records, reliably improving long-horizon planning 18% through simulation, debugging, reproducible testing, performance analysis, and iterative system optimization at scale.
- Analyzed time-sensitive system interactions and edge cases, increasing closed-loop stability 15% while documenting implementation tradeoffs, experimental findings, model behavior, and reproducible technical decisions for senior researchers at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Java and Python microservices using REST APIs, Docker, SQL, AWS Glue, and S3, improving response times 30%, reducing payload sizes 25%, and lowering storage costs 15%.
- Automated 20+ unit and integration tests across Windows, macOS, and Linux while increasing pipeline efficiency 68% and collaborating with Microsoft and GitHub stakeholders during Agile reviews and deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built eight object-oriented Node.js services for authentication, resource allocation, customer records, billing, permissions, and notifications, reliably synchronizing two million records through WebSockets, REST APIs, and Azure SQL.
- Automated testing and Docker-based deployments through Jenkins, reducing release time 50% while using Sentry, Azure Monitor, Kibana, and operational analytics to improve reliability across production release cycles.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, Shell, Bash
Systems: OOP, data structures, algorithms, REST APIs, distributed systems, microservices, WebSockets, caching, fault tolerance
Data/Cloud: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, AWS, Azure, Docker, Kubernetes
Quality: Git, CI/CD, Jenkins, PyTest, Jest, unit testing, integration testing, debugging, monitoring, observability

Projects

CPU-scheduling Simulator Live | *Reliable Systems*
- Implemented six C++ scheduling algorithms with edge-case handling, automated waiting and turnaround metrics, and Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25%.

FinNexus Live | *Real-Time Distributed Platform*
- Built a distributed platform reliably supporting 50,000+ users and one million transactions using TypeScript, MongoDB, WebSockets, Kubernetes, secure access controls, real-time updates, filtering, and automated deployment workflows.

Chat2SQL Code | *APIs, Databases, Integration*
- Developed Java and TypeScript services supporting 10+ databases, configuring Docker workflows across Windows, macOS, and Linux with schema synchronization, structured validation, reliable local debugging, and cross-platform deployment.